

Lecture 9

Bilinear Groups and BGN

SEC 595: Encrypted Computing

Roadmap

Lecture 8 gave partially homomorphic schemes: ElGamal multiplies and Paillier adds, each supporting *one* operation an unlimited number of times. This lecture gives the first somewhat homomorphic scheme of the course, due to Boneh, Goh and Nissim (2005): BGN supports any number of additions and exactly *one* multiplication. Section 1 introduces the object that supplies the multiplication, a bilinear pairing. Section 2 builds BGN in a group of composite order $n = q_1 q_2$ and decrypts it. Section 3 gives the additive homomorphism, Section 4 the single multiplication and why there is only one, and Section 5 the security assumption.

1. Bilinear Groups

1.1 Bilinear Maps

Definition 1 (Bilinear map). Let $\mathbb{G}$ and $\mathbb{G}_1$ be multiplicative cyclic groups of order n , and let g generate $\mathbb{G}$. A map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_1$ is **bilinear** if

$$e(u^a, v^b) = e(u, v)^{ab} \quad \text{for all } u, v \in \mathbb{G}, a, b \in \mathbb{Z},$$

and **non-degenerate** if $e(g, g)$ generates $\mathbb{G}_1$. $\mathbb{G}$ is a **bilinear group** if such a $\mathbb{G}_1$ and e exist.

Both arguments come from the same group, so the pairing is *symmetric*: a ciphertext can be paired with a ciphertext. Efficient constructions exist on elliptic curves (Lecture 1); everything below uses only the two defining properties. In composite order, asking that $e(g, g)$ generate $\mathbb{G}_1$ is stronger than asking $e(g, g) \neq 1$: a non-trivial element may still have order a proper divisor of n .

1.2 What the Pairing Buys, and What It Costs

Setting $u = v = g$,

$$e(g^a, g^b) = e(g, g)^{ab}.$$

Two hidden exponents go in and their *product* comes out, without a or b ever being exposed. This happens once: the result lies in $\mathbb{G}_1$, and there is no map out of $\mathbb{G}_1$.

The price is that DDH becomes easy in $\mathbb{G}$.

Proposition 1. *Given $g, g^a, g^b, z \in \mathbb{G}$, one can decide whether $z = g^{ab}$.*

Proof. Test $e(g^a, g^b) \stackrel{?}{=} e(g, z)$. With $z = g^c$ the two sides are $e(g, g)^{ab}$ and $e(g, g)^c$; since $e(g, g)$ has order n , they are equal iff $c \equiv ab \pmod{n}$. $\square$

So BGN cannot rely on DDH. It relies instead on the composite order of $\mathbb{G}$ (Section 5).

Example 1 (Checking bilinearity). Let $\mathbb{G} = \langle g \rangle$.

(a) Write $u_1 = g^{a_1}$, $u_2 = g^{a_2}$, $v = g^b$. Then

$$e(u_1 u_2, v) = e(g^{a_1+a_2}, g^b) = e(g, g)^{(a_1+a_2)b} = e(g, g)^{a_1 b} e(g, g)^{a_2 b} = e(u_1, v) e(u_2, v).$$

So in a cyclic group the exponent form of bilinearity implies the usual additive form.

(b) $e(g^a, g^b) e(g^c, g^d) = e(g, g)^{ab} e(g, g)^{cd} = e(g, g)^{ab+cd}$. A product of pairings computes a sum of products in the exponent; Section 4.4 uses exactly this.

2. BGN: Keys, Encryption, Decryption

2.1 The Composite-Order Group

A generator $\mathcal{G}(\tau)$, on security parameter τ , outputs $(q_1, q_2, \mathbb{G}, \mathbb{G}_1, e)$ with

$$|\mathbb{G}| = |\mathbb{G}_1| = n = q_1 q_2,$$

where q_1, q_2 are distinct primes of similar size and e is bilinear.

Proposition 2. *Let u generate $\mathbb{G}$ and $h = u^{q_2}$. Then $\text{ord}(h) = q_1$, and $h = g^{\alpha q_2}$ for some α with $\gcd(\alpha, q_1) = 1$.*

Proof. In a cyclic group of order n , $\text{ord}(u^k) = n / \gcd(n, k)$; with $k = q_2$ this is $n / q_2 = q_1$. If $h = g^t$, then $n / \gcd(n, t) = q_1$ forces $\gcd(n, t) = q_2$, so $t = \alpha q_2$ and $\gcd(q_1 q_2, \alpha q_2) = q_2 \gcd(q_1, \alpha) = q_2$ gives $\gcd(\alpha, q_1) = 1$. $\square$

The exponent α is never published; it appears only in the algebra of Section 4.

2.2 The Scheme

Definition 2 (BGN). • **KeyGen**(τ): run $\mathcal{G}(\tau)$ to obtain $n = q_1 q_2$. Pick random generators $g, u \xleftarrow{R} \mathbb{G}$ and set $h = u^{q_2}$. Output

$$\text{Pk} = (n, \mathbb{G}, \mathbb{G}_1, e, g, h), \quad \text{Sk} = q_1.$$

• **Enc**(Pk, m), $m \in \{0, 1, \dots, T\}$ with $T < q_2$: pick $r \xleftarrow{R} \{0, \dots, n-1\}$ and output

$$C = g^m h^r \in \mathbb{G}.$$

• **Dec**(Sk, C): compute C^{q_1} and output $m = \log_{g^{q_1}} C^{q_1}$.

The four objects have separate jobs:

g , of order n	carries the message
h , of order q_1	carries the mask
q_1	annihilates the mask
q_2	bounds the message space

The public key contains n , and Sk is one of its factors, so whoever factors n holds the secret key.

2.3 Correctness

Proposition 3. *For every r , $(h^r)^{q_1} = 1$, so $C^{q_1} = (g^{q_1})^m$. Moreover $\text{ord}(g^{q_1}) = q_2$, so Dec returns $m \bmod q_2$, and is correct exactly when $T < q_2$.*

Proof. $(h^r)^{q_1} = (u^{q_1 q_2})^r = (u^n)^r = 1$. Next, $\text{ord}(g^{q_1}) = n / \gcd(n, q_1) = q_2$, so $(g^{q_1})^m$ depends only on $m \bmod q_2$, and is injective on $\{0, \dots, T\}$ iff $T + 1 \leq q_2$. $\square$

The mask is cancelled exactly, whatever r was: there is no noise. But the message is still in an exponent, so decryption ends with a discrete logarithm. It is feasible only because m is tiny. With $T = 1$ it is a single comparison: is C^{q_1} equal to 1 or to g^{q_1} ? For T near q_2 it is a full discrete logarithm and infeasible. This is a condition on the cost of decryption, not on security; in practice BGN encrypts bits or very small integers.

Example 2 (A toy instance). Take $q_1 = 5$, $q_2 = 7$, $n = 35$. Since $35 \mid 70 = 71 - 1$, $\mathbb{Z}_{71}^*$ has a unique subgroup of order 35; let $\mathbb{G}$ be that subgroup. Take $g = 2$ and $u = 27$, both of order 35 modulo 71.

Keys. $h = u^{q_2} = 27^7 \bmod 71 = 25$, and $25^5 \equiv 1$ with $\text{ord}(25) = 5 = q_1$. Since $2^{21} \equiv 25 \pmod{71}$, $\alpha q_2 = 21$ and $\alpha = 3$, coprime to $q_1 = 5$. So

$$\text{Pk} = (35, \mathbb{G}, \mathbb{G}_1, e, 2, 25), \quad \text{Sk} = 5, \quad T = 1 < 7 = q_2.$$

Encrypt each bit with $r = 6$ and $r = 13$, computing $C = 2^m 25^r \bmod 71$:

m	r	C
0	6	25
0	13	5
1	6	50
1	13	10

The four ciphertexts are distinct; nothing in the list shows which two encrypt 0.

Example 3 (Decrypting the toy ciphertexts). Here $g^{q_1} = 2^5 = 32$, and its powers modulo 71 are

$$32^0, \dots, 32^6 = 1, 32, 30, 37, 48, 45, 20,$$

seven distinct values with $32^7 \equiv 1$, so $\text{ord}(32) = 7 = q_2$. Raising the ciphertexts to $q_1 = 5$:

$$25^5 \equiv 1, \quad 5^5 \equiv 1, \quad 50^5 \equiv 32, \quad 10^5 \equiv 32 \pmod{71}.$$

So 25 and 5 decrypt to 0, and 50 and 10 decrypt to 1. $\checkmark$

3. Additive Homomorphism

Proposition 4. *If $C_i = g^{m_i} h^{r_i}$, then*

$$C_1 C_2 = g^{m_1 + m_2} h^{r_1 + r_2},$$

a ciphertext of $m_1 + m_2$, which decrypts to $m_1 + m_2 \bmod q_2$.

Proof. $\mathbb{G}$ is abelian, so the factors regroup; decryption is Proposition 3. $\square$

The product has the same form as a fresh ciphertext and is decrypted by the same q_1 . Since the mask is cancelled exactly rather than accumulated, there is no budget: additions chain without limit. What is limited is the *plaintext*, which must stay below q_2 .

Example 4 (Addition and its limit). In the toy instance, $C_1 = \text{Enc}(1; r = 6) = 50$ and $C_2 = \text{Enc}(1; r = 3) = 2 \cdot 25^3 \bmod 71 = 10$. Then

$$C_1 C_2 = 500 \equiv 3 \pmod{71}, \quad 3 = 2^2 \cdot 25^9 \bmod 71,$$

as Proposition 4 predicts. Decrypting: $3^5 = 243 \equiv 30 \pmod{71}$, and $30 = 32^2$, so the result is 2.

With $T = 1$ the value 2 is not a bit: it is the correct *integer* sum $1 + 1$, not $1 \oplus 1$, and the caller must read it that way. Pushing on, a product of seven encryptions of 1 has $C^{q_1} = 32^7 = 1$ and decrypts to $7 \bmod 7 = 0$: the wraparound of Proposition 3 becomes a wrong answer.

4. The One Multiplication

4.1 The Target Group

Two elements of $\mathbb{G}_1$ are computable from Pk:

$$g_1 = e(g, g), \quad h_1 = e(g, h) = e(g, g^{\alpha q_2}) = g_1^{\alpha q_2}.$$

By non-degeneracy $\text{ord}(g_1) = n$, and as in Proposition 2, $\text{ord}(h_1) = n / \gcd(n, \alpha q_2) = q_1$. So $\mathbb{G}_1$ has the same message/mask structure as $\mathbb{G}$, with (g_1, h_1) in place of (g, h) .

4.2 Homomorphic Multiplication

Definition 3 (Multiplication). Given $C_1, C_2 \in \mathbb{G}$, pick a fresh $r \in \mathbb{Z}_n$ and output

$$C = e(C_1, C_2) h_1^r \in \mathbb{G}_1.$$

Proposition 5. If $C_i = g^{m_i} h^{r_i}$, then $C = g_1^{m_1 m_2} h_1^{\tilde{r}}$ with

$$\tilde{r} = m_1 r_2 + m_2 r_1 + \alpha q_2 r_1 r_2 + r.$$

Proof. With $h = g^{\alpha q_2}$, $C_i = g^{m_i + \alpha q_2 r_i}$, and bilinearity gives

$$e(C_1, C_2) = g_1^{(m_1 + \alpha q_2 r_1)(m_2 + \alpha q_2 r_2)} = g_1^{m_1 m_2} g_1^{\alpha q_2 (m_1 r_2 + m_2 r_1 + \alpha q_2 r_1 r_2)}.$$

Since $g_1^{\alpha q_2} = h_1$, this is $g_1^{m_1 m_2} h_1^{m_1 r_2 + m_2 r_1 + \alpha q_2 r_1 r_2}$; multiplying by h_1^r adds r to the exponent of h_1 . $\square$

Every term other than $m_1 m_2$ carries a factor αq_2 and so lands in $\langle h_1 \rangle$. Evaluation uses only public data; $\tilde{r}$ is never computed, since decryption needs only that the second factor lies in the subgroup of order q_1 .

4.3 Decrypting in $\mathbb{G}_1$

Proposition 6. For $C = g_1^{m_1 m_2} h_1^{\tilde{r}}$, $C^{q_1} = (g_1^{q_1})^{m_1 m_2}$ and $\text{ord}(g_1^{q_1}) = q_2$, so $m_1 m_2 = \log_{g_1^{q_1}} C^{q_1}$ is recovered by search when $m_1 m_2 < q_2$.

Proof. $h_1^{\tilde{r} q_1} = (h_1^{q_1})^{\tilde{r}} = 1$ since $\text{ord}(h_1) = q_1$; and $\text{ord}(g_1^{q_1}) = n/q_1 = q_2$. The rest is Proposition 3. $\square$

The secret key is the same q_1 ; only the base of the logarithm changes, from g^{q_1} to $g_1^{q_1}$, so the decryptor must know which group C lies in.

4.4 Why Exactly One Multiplication

The inputs live in $\mathbb{G}$; the product lives in $\mathbb{G}_1$. There is no map out of $\mathbb{G}_1$, so a product ciphertext cannot be paired again. BGN's ceiling is therefore a *typing* limit, not a noise bound: larger q_1, q_2 do not raise it.

Product ciphertexts still have the form $g_1^m h_1^{\tilde{r}}$, so Proposition 4 applies in $\mathbb{G}_1$: multiplying them adds their messages. The supported functions are **sums of products of pairs** — polynomials of total degree at most 2, with any number of terms, whose value stays below q_2 .

Example 5 (Expanding the exponent). (a) Using $h = g^{\alpha q_2}$ and bilinearity,

$$e(g^{m_1} h^{r_1}, g^{m_2} h^{r_2}) h_1^r = g_1^{m_1 m_2 + \alpha q_2 m_1 r_2 + \alpha q_2 m_2 r_1 + \alpha^2 q_2^2 r_1 r_2} h_1^r.$$

(b) Replacing $g_1^{\alpha q_2}$ by h_1 , this is $g_1^{m_1 m_2} h_1^{\tilde{r}}$ with $\tilde{r} = m_1 r_2 + m_2 r_1 + \alpha q_2 r_1 r_2 + r$.

(c) $h_1^{\tilde{r} q_1} = (h_1^{q_1})^{\tilde{r}} = 1^{\tilde{r}} = 1$, since $\text{ord}(h_1) = q_1$; equivalently, $h_1^{\tilde{r} q_1} = g_1^{\alpha \tilde{r} q_1 q_2} = (g_1^n)^{\alpha \tilde{r}} = 1$.

Example 6 (A multiplication in the toy instance). *Model.* Let $\mathbb{G}_1$ be the subgroup of order 35 in $\mathbb{Z}_{211}^*$ (since $35 \mid 210$), with generator $g_1 = 5$, and define $e(g^a, g^b) = g_1^{ab \bmod 35}$. This map is bilinear and non-degenerate, but not efficient: evaluating it needs discrete logarithms in $\mathbb{G}$. It is exact for checking the algebra.

(a) $h_1 = g_1^{\alpha q_2} = 5^{21} \bmod 211 = 107$, of order $5 = q_1$; and $g_1^{q_1} = 5^5 \bmod 211 = 171$, of order $7 = q_2$.

(b) Take $C_1 = 50$ ($m_1 = 1, r_1 = 6$) and $C_2 = 10$ ($m_2 = 1, r_2 = 3$). As powers of g : $1 + 21 \cdot 6 = 127 \equiv 22$ and $1 + 21 \cdot 3 = 64 \equiv 29 \pmod{35}$, so $C_1 = g^{22}, C_2 = g^{29}$. With $r = 11$,

$$e(C_1, C_2) = g_1^{22 \cdot 29 \bmod 35} = g_1^8, \quad h_1^{11} = g_1^{21 \cdot 11 \bmod 35} = g_1^{21},$$

so $C = g_1^{8+21} = g_1^{29} = 5^{29} \bmod 211 = 96$.

(c) *Decrypt:* $C^{q_1} = 96^5 \bmod 211 = 171 = (g_1^{q_1})^1$, so the result is $1 = m_1 m_2$. $\checkmark$ *Check* against Proposition 5:

$$\tilde{r} = 1 \cdot 3 + 1 \cdot 6 + 3 \cdot 7 \cdot 6 \cdot 3 + 11 = 3 + 6 + 378 + 11 = 398,$$

and $g_1^1 h_1^{398} = g_1^{1+21 \cdot 398 \bmod 35} = g_1^{29} = 96$. $\checkmark$

Example 7 (What BGN can compute). On encrypted bits $x_1, \dots, x_k$ and $y_1, \dots, y_k$:

- (a) $x_1 + \dots + x_k$: yes, by multiplying ciphertexts in $\mathbb{G}$; correct if $k < q_2$.
- (b) $x_1 x_2 + x_3 x_4$: yes, two pairings, then one multiplication in $\mathbb{G}_1$; correct since $2 < q_2$.
- (c) $\sum_i x_i y_i$: yes, k pairings multiplied together in $\mathbb{G}_1$; correct if $k < q_2$.

(d) $x_1x_2x_3$: no. It has degree 3, and $e(C_1, C_2)$ lies in $\mathbb{G}_1$, where it cannot be paired with C_3 .

In each accepted case the condition is that the largest possible value of the result is below q_2 , so that decryption returns the integer rather than a residue.

5. Security: Subgroup Decision

Definition 4 (Subgroup decision assumption). Given $(n, \mathbb{G}, \mathbb{G}_1, e)$ but not q_1, q_2 , no efficient algorithm distinguishes a uniform element of $\mathbb{G}$ from a uniform element of the subgroup of order q_1 with more than negligible advantage.

Proposition 7. *Under the subgroup decision assumption, BGN is IND-CPA secure (Lecture 8, §1.5).*

Proof sketch. Replace h in Pk by a uniform element of $\mathbb{G}$; by the assumption the attacker's advantage changes negligibly. A uniform h generates $\mathbb{G}$ with high probability, and then h^r is uniform in $\mathbb{G}$, so $C = g^m h^r$ is uniform and independent of m ; the advantage in this game is 0. $\square$

Factoring n breaks the assumption (raise a candidate to the power q_1 and test for 1), so BGN is at most as hard as factoring n . DDH is not available as an alternative, since the pairing breaks it (Section 1.2).

Summary

1. A pairing $e(u^a, v^b) = e(u, v)^{ab}$ multiplies hidden exponents once; it also makes DDH easy in $\mathbb{G}$.
2. BGN works in a group of order $n = q_1 q_2$: $h = u^{q_2}$ has order q_1 , $C = g^m h^r$, Pk = $(n, \mathbb{G}, \mathbb{G}_1, e, g, h)$, Sk = q_1 .
3. $C^{q_1} = (g^{q_1})^m$ removes the mask exactly; a small discrete logarithm then recovers m , so $T < q_2$ and in practice $T = 1$.
4. $C_1 C_2$ adds messages without limit; $e(C_1, C_2) h_1^r$ multiplies them once, landing in $\mathbb{G}_1$. BGN evaluates polynomials of degree at most 2, a typing limit rather than a noise limit.
5. Security rests on subgroup decision, which is at most as hard as factoring n .

Exercises

1. In the model of Example 6, verify $e(g^3, g^4) e(g^2, g^5) = g_1^{22} = 113 \pmod{211}$, using Example 1(b).
2. Repeat the key generation of Example 2 with $u = 16$. Compute h , check $\text{ord}(h) = 5$, find α , and confirm $\gcd(\alpha, q_1) = 1$. Encrypt $m = 1$ with $r = 6$ and decrypt with Sk = 5.
3. State exactly which pairs $m, m' \geq 0$ decrypt to the same value in BGN. In the toy instance, what does a product of nine encryptions of 1 decrypt to?
4. In the toy instance, multiply $C_1 = 25$ (an encryption of 0 with $r_1 = 6$) and $C_2 = 10$ (an encryption of 1 with $r_2 = 3$) using $r = 11$. Compute C , decrypt it, and check your C against $g_1^{m_1 m_2} h_1^{\tilde{r}}$ with $\tilde{r}$ from Proposition 5.

5. Write BGN in the interface of Lecture 8: give KeyGen, Enc, Dec and Evaluate for $\Pi = +$ and $\Pi = \times$. Is BGN compact? Justify.
6. Show how BGN computes the number of indices i with $x_i = y_i$ for two encrypted bit vectors of length k . (Hint: $[x = y] = 1 - x - y + 2xy$; how are the constant and the negative signs handled modulo q_2 ?) What condition on k and q_2 is needed?
7. Explain in two sentences why choosing larger q_1, q_2 cannot give BGN a second multiplication, and contrast this with the integer scheme $c = pq + 2r + m$ of Lecture 8, whose depth grows with p .